



**Lab Sheet-01**  
**BCA201P**  
**Foundations of Java Development**  
**BCA III Semester (Session 2026-2027)**

**Objective:** To understand the basic concepts of variable, decision and loop control statements.

**Requirements:**

- VS Code editor or eclipse or IntelliJ Idea IDE
- JDK 21 or higher

**Assignments:**

1. Write a Java program to display your name, age, and college name using variables of appropriate data types.
2. Input two integers and print the result of all arithmetic operations (addition, subtraction, multiplication, division, and modulus).
3. Write a Java program to check whether a given number is even or odd using the modulus operator and if-else.
4. Write a program to demonstrate the use of relational operators by comparing two numbers.
5. Accept a floating-point number and convert it explicitly to an integer. Print both the original and the converted values.
6. Write a program to accept a character and print its ASCII value.
7. Input a number and check if it is divisible by both 3 and 5 using logical operators.
8. Input a character and check whether it is a vowel or a consonant using if-else.
9. Input marks of three subjects, calculate total and percentage, and print pass/fail using if-else (passing criteria:  $\geq 40\%$  in all subjects).
10. Write a program to find the greatest of two numbers using if-else.



### Intermediate Level (11–20)

11. Input three numbers and print the largest using nested if-else.
12. Accept a year and check whether it is a leap year using if-else.
13. Accept a number and check whether it is positive, negative, or zero using if-else.
14. Accept a character and check if it is a digit, uppercase letter, lowercase letter, or a special character using character classification and if-else.
15. Accept a 3-digit number and check whether it is an Armstrong number using arithmetic operators and if-else.
16. Input salary and calculate tax based on income slabs using if-else:
  - Income  $\leq$  250000: No tax
  - 250001 to 500000: 5%
  - 500001 to 1000000: 20%
  - Above 1000000: 30%
17. Accept an integer and check whether it is a palindrome using arithmetic operations and type casting.
18. Accept a number and print whether it is prime or not using if-else.
19. Accept a character and use if-else to identify whether it's a valid operator (+, -, \*, /) and perform the corresponding operation on two input numbers.
20. Accept age and gender and determine eligibility for marriage (male  $\geq$  21, female  $\geq$  18) using nested if-else conditions.

### Advanced Level (21–30)

21. Accept three angles and check whether they can form a valid triangle (sum of angles =  $180^\circ$ ).
22. Input the length and breadth of a rectangle, and determine whether it is a square using if-else.
23. Input a temperature in Celsius and convert it to Fahrenheit using type casting and expressions.
24. Accept a number and check whether it is a perfect number (sum of divisors = number) using basic arithmetic and if-else.
25. Accept two integers and a logical operator as a character. Use if-else to perform the logical operation and display the result.
26. Input a five-digit number and check whether the first and last digits are the same using arithmetic expressions.
27. Input the cost price and selling price of an item. Determine profit or loss, and by how much, using if-else.
28. Accept a number and display the number of digits using expressions and if-else.



29. Input the height (in cm) of a person and classify them as short ( $<150$ ), average ( $150-170$ ), or tall ( $>170$ ).
30. Write a Java program to simulate a simple calculator that takes two numbers and an operator (+, -, \*, /) and performs the corresponding operation using if-else.

### Instructions:

- Submit both the source code and the corresponding output.
- **Handwrite your lab sheets in a spiral notebook.**
- Begin each new program on a fresh page.
- Before submission, carefully review your code for errors or possible improvements.  
Make the necessary corrections or enhancements before finalizing your work.

